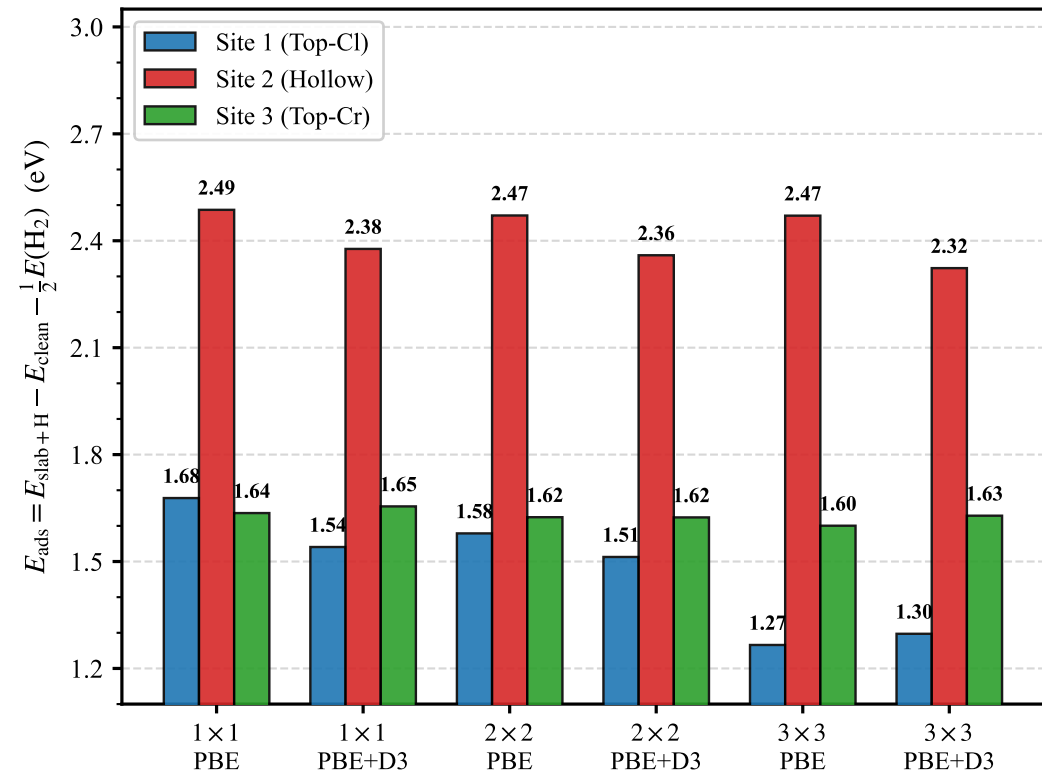
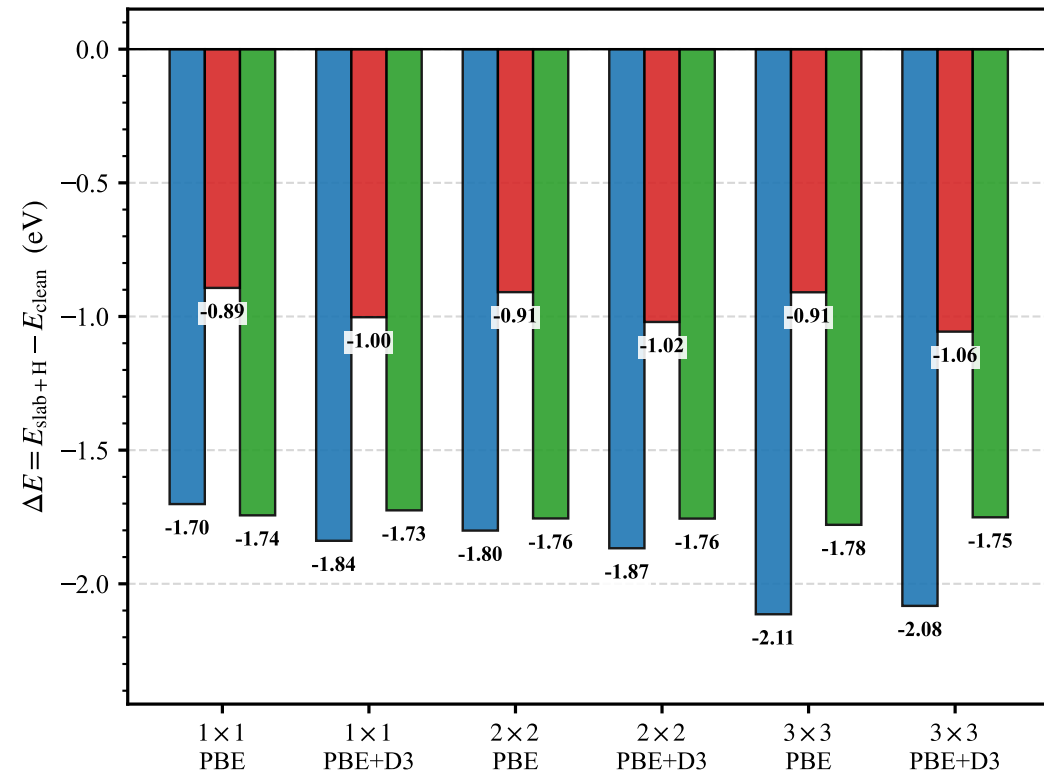


Multi-Scale Hydrogen Adsorption Thermodynamics on Monolayer CrCl_3 : 1×1 vs 2×2 vs 3×3

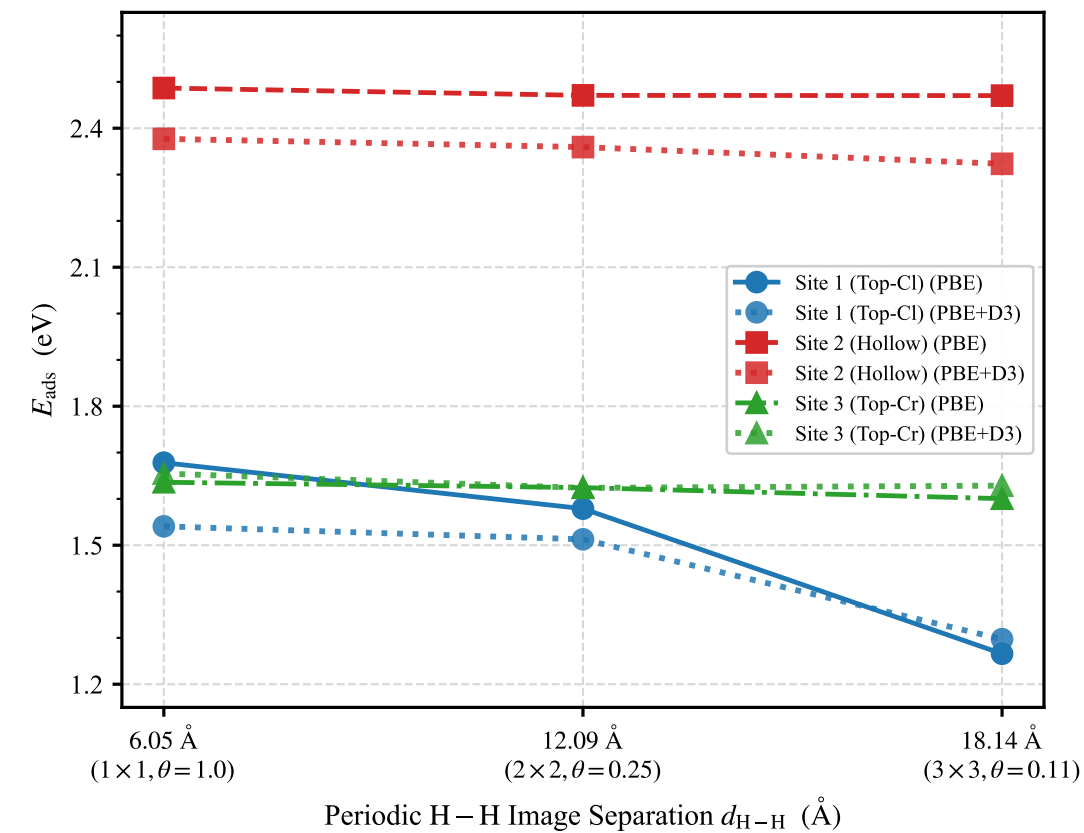
(a) Hydrogen Adsorption Energy (1×1 vs 2×2 vs 3×3)



(b) Thermodynamic Binding Energy ΔE



(c) Multi-Scale Coverage Convergence: E_{ads} vs $d_{\text{H-H}}$



(d) HER Free Energy Profile ($\Delta G_{\text{H}^*} = E_{\text{ads}} + 0.24$ eV)

